

# Smart Healthcare Monitoring & AI Consultation Platform

## Project Design Report (PDR)

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### 1. Introduction

The Smart Healthcare Monitoring & AI Consultation Platform is a modern healthcare ecosystem designed to simplify personal health management through AI-assisted recommendations, emergency support systems, and smart monitoring features.

### 2. Objectives

The project aims to provide centralized healthcare management, AI-based suggestions, doctor appointment systems, medicine ordering, emergency healthcare support, and interactive health monitoring.

### 3. Health Monitoring Dashboard

The dashboard provides weight tracking, heart rate monitoring, ECG animations, health suggestions, reminder notifications, and charts for better healthcare visualization.

### 4. Add Health Records

Users can add weight, heart rate, symptoms, and health notes. Records can be edited, deleted, and displayed in timeline format.

### 5. AI-Based Health Suggestions

The system analyzes symptoms and recommends precautions and doctor specialists. Example: Fever → General Physician, Chest Pain → Cardiologist.

### 6. Doctor Appointment System

Users can choose doctors, select online/offline consultations, pick dates and time slots, and book appointments with confirmation.

## 7. Medicine Ordering System

The application includes medicine browsing, cart management, address input, order placement, and medicine delivery simulation.

## 8. Emergency Support System

The emergency module provides ambulance support, nearby hospital access, emergency contacts, and future emergency detection logic.

## 9. Medical Timeline & Reports

Users can upload prescriptions and maintain medical history timelines for better healthcare organization.

## 10. Technologies Used

Frontend: React.js, React Router, CSS3, Recharts. Backend: Node.js, Express.js. Database: MongoDB Atlas.

## 11. Future Enhancements

Future upgrades include smartwatch integration, live health tracking, Google Maps hospital finder, AI chatbot integration, and video consultation support.

## 12. Conclusion

The Smart Healthcare Platform combines healthcare monitoring, AI recommendations, emergency support, and medical management into a single interactive ecosystem.

## 13. Project Modules Summary

| Module              | Description                                       |
|---------------------|---------------------------------------------------|
| Dashboard           | Health statistics and ECG animations              |
| Doctor Consultation | Appointment booking and specialist recommendation |

|                   |                                   |
|-------------------|-----------------------------------|
| Medicine Ordering | Cart and medicine delivery system |
| Emergency Support | Emergency healthcare assistance   |
| AI Suggestions    | Symptom-based healthcare guidance |